

Model Failure Challenge Teacher Notes

TOK Cognitive Science Activity Bank • Natural Sciences • 25-35 min

Category	Natural Sciences	Time	25-35 min
Difficulty	Medium	ID	model-failure-challenge

Big idea: Models simplify reality productively.

Overview

Students learn that models are useful simplifications rather than perfect copies of reality.

Student Prompt

Inspect Model Failure Challenge Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Model Failure Challenge Stimulus A	Student-facing stimulus 1: A simple weather, disease, or traffic model that works until one assumption changes. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Model Failure Challenge Stimulus B	Student-facing stimulus 2: An anomaly card that forces students to revise a model. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Model Failure Challenge Reveal / Twist	Student-facing stimulus 3: A model limits table: useful for, weak when, assumptions, and missing variables. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Models simplify reality productively.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Model Failure Challenge Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Model Failure Challenge Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Are models useful because they are false?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how models simplify reality productively.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Prepare a starter stimulus using: A simple weather, disease, or traffic model that works until one assumption changes.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Ready-to-Use Examples

- A simple weather, disease, or traffic model that works until one assumption changes.
- An anomaly card that forces students to revise a model.
- A model limits table: useful for, weak when, assumptions, and missing variables.

Suggested Timing

Stage	Teacher move	Watch for
1	Groups identify variables missing from the model: prior knowledge, sleep, stress, revision quality, teaching, test difficulty.	Assumptions, confidence shifts, evidence claims, and language choices.
2	They draw a revised model with extra variables.	Assumptions, confidence shifts, evidence claims, and language choices.
3	They decide whether the original model is useless, useful or dangerous.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Discuss when a model should be rejected.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Common Misconceptions

- Students may think observation happens before theory in a simple linear order.
- Students may overclaim from a classroom model without naming its limits.

Assessment Look-Fors

- Names variables, assumptions, controls, anomalies, and limits.
- Explains how methods shape what can count as scientific knowledge.

Differentiation and Adaptations

- Short lesson: use only the first example, one response grid, and one debrief question.
- Long lesson: add a second round where students revise the method and compare outcomes.
- Low-tech option: run with printed cards, sticky notes, and a board tally.

Extension Tasks

- Design a second version of the activity that tests the same knowledge issue in another AOK.
- Find a real-world example where the same knowledge problem matters outside the classroom.
- Write a 150-word TOK exhibition-style commentary using the activity as the object context.

Related Activities

- Observation is Theory-Laden
- P-hacking with Random Data
- Placebo Trial Design